

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The tcpdump log analysis indicates a failure in DNS communication. When attempting to access www.yummyrecipesforme.com, the browser sent DNS queries using the **UDP protocol** to a DNS server in order to obtain the website's IP address. Instead of receiving a valid DNS response, the system received **ICMP error messages** stating:

"udp port 53 unreachable"

The network traffic involved the following protocols:

- **DNS (Domain Name System)** – Used to translate the domain name into an IP address.
- **UDP (User Datagram Protocol)** – Used to transmit DNS queries.
- **ICMP (Internet Control Message Protocol)** – Used to return error messages when network communication fails.

Analysis of the log revealed:

- Multiple DNS requests were sent from **192.51.100.15** to the DNS server **203.0.113.2**.
- Each request attempted to resolve the domain **yummyrecipesforme.com**.
- The query contained the **A?** flag, indicating a request for an IPv4 address record.
- Every DNS request received an ICMP response indicating that **UDP port 53 was unreachable**.
- Port **53** is the standard port used by DNS services.

Based on these findings, the DNS server was unable to process requests on port 53. As a result, domain name resolution failed and users could not access the website. The evidence suggests that the DNS service was unavailable,

misconfigured, or blocked from receiving traffic.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

Time Incident Occurred

The first recorded event in the tcpdump log occurred at:

13:24:32.192571 (1:24 PM)

Scenario, Events, and Symptoms

Customers reported that they were unable to access www.yummyrecipesforme.com and received a "**Destination Port Unreachable**" error message after attempting to load the webpage.

To investigate the issue, network traffic was captured using **tcpdump** while attempting to access the website. The packet capture showed DNS requests being sent successfully but receiving ICMP error responses instead of DNS replies.

Current Status

The incident has been reported to management and escalated to the security engineering team. Investigation and remediation efforts are currently in progress to restore website accessibility.

Information Discovered During Investigation

The investigation identified the following:

- DNS requests were transmitted using UDP.
- The requests targeted DNS service on port 53.
- The DNS server repeatedly returned ICMP error messages indicating "**udp port 53 unreachable.**"
- No successful DNS responses were observed.
- The website could not be reached because the domain name could not be resolved to an IP address.

Next Steps

1. Verify that the DNS server is operational.
2. Confirm that the DNS service is listening on UDP port 53.
3. Review firewall rules to ensure DNS traffic is not blocked.
4. Examine DNS server logs for errors or service failures.
5. Investigate recent configuration changes.
6. Assess whether a Denial-of-Service (DoS) attack affected the DNS server.

Suspected Root Cause

The most likely causes of the incident include:

- DNS server outage or service failure.
- Firewall rules blocking DNS traffic on port 53.
- DNS server misconfiguration.
- A potential Denial-of-Service (DoS) attack that disrupted DNS availability.

Based on the repeated ICMP error responses and the inability to resolve the domain name, the incident appears to be caused by the DNS server being unavailable or inaccessible through port 53.